

Lim, Hyeon Jun

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EDUCATION

- **Yonsei University** Seoul, South Korea
M.S. in Electrical and Electronic Engineering *Feb. 2018*
 - Thesis: “State Estimation Using the Sampled-data Fuzzy Observer and Its Application to the AHRS”
 - BK21 Plus Scholarship, National Research Foundation of Korea (NRF)
- **Yonsei University** Seoul, South Korea
B.S. in Electrical and Electronic Engineering *Feb. 2016*
 - National Science and Engineering Scholarship, Government of South Korea

PROFESSIONAL EXPERIENCE SUMMARY

- Senior Research Engineer with 7+ years of experience in robotics and autonomous driving, specializing in Reinforcement Learning (RL), Transformer-based End-to-End (E2E) planning, and high-reliability decision-making systems. Experienced in architecting closed-loop simulation environments, developing LiDAR-based perception and 3D reconstruction systems, and integrating real-vehicle software for autonomous driving platforms. Built and managed large-scale autonomous-driving datasets exceeding 1M labeled objects, and designed LiDAR simulation frameworks supporting sensor-technology development, HD mapping, and high-fidelity 3D scanning. Strong theoretical foundation in T-S fuzzy-based robust control, complemented by hands-on experience delivering scalable, safety-critical algorithms and infrastructure across both research and industrial environments.

EMPLOYMENT

- **Hyundai Motor Company** Seoul, South Korea
Senior Research Engineer, Autonomous Driving Development *Jan. 2025 – Present*
- **Hyundai Motor Company** Seoul, South Korea
Research Engineer, Autonomous Driving Development *Mar. 2022 – Dec. 2024*
- **Korea Institute of Industrial Technology (KITECH)** Ansan, South Korea
Researcher, Robotics R&BD Group *Jun. 2018 – Jun. 2021*

KEY PROJECTS

- **Unified Closed-Loop Simulation Framework for Reinforcement Learning-Based Autonomous Driving** *Aug. 2025 – Present*
 - Architecting a high-fidelity closed-loop simulation stack across several environment representations, from 3DGS photorealistic scenes and graphics-based simulation to object-level box worlds, with rule-driven reward modeling and World-Model-powered synthetic data generation, to enable reactive long-tail scenario evaluation and robust Sim-to-Real generalization.
 - Optimizing real-world driving policies by integrating online and offline RL paradigms and designing specialized neural layers for transfer.
- **End-to-End Autonomous Planning System** *Jul. 2024 – Mar. 2026*
 - Co-developed a Transformer-based autonomous-driving planner and led its embedded deployment and real-vehicle validation, including the company’s first on-vehicle run of such a planner. Built a unified training and inference framework, architected closed-loop evaluation and long-tail robustness analysis with high-fidelity synthetic data, and accelerated iteration through custom real-time visualization and debugging tools used across diverse real-vehicle tests.
- **Lv.3 and Lv.4 Autonomous Driving Perception Systems (HDP & Robo-Services)** *Mar. 2022 – Jun. 2024*

- Shaped the software architecture of the company's LiDAR perception stack and led sensor evaluation and point cloud pre-processing, establishing the first systematic sensor-evaluation framework and benchmarking 10+ vendors. Contributed core parsing, filtering, fault-detection, clustering, localization, and calibration technologies, including online optical-flow calibration, with CUDA-optimized embedded performance, validated across Lv.3 and Lv.4 platforms.
- **Standardized Open Database and Evaluation System for Intelligent Vehicle Perception** *Jun. 2018 – May 2020*
 - Built the Korea Dataset for Autonomous Solution (KODAS), the country's first large-scale, high-precision LiDAR dataset for autonomous driving, constructing a million-label 128-channel corpus with automated annotation (90% time reduction), long-tail extreme-condition captures, cm-level HD mapping with custom tooling (LAS converter, point-wise sync, SOR filtering, rolling-shutter correction), and a full-stack calibrated sensor-logging platform.
 - Advanced 3D perception fidelity and validation rigor by boosting Z-axis accuracy from 10 cm to 1 cm via global ground-fitting, performing comprehensive multi-sensor calibration, and benchmarking SOTA deep-learning models to confirm real-world applicability and robustness of the dataset.
- **River Management Technology Using a Drone-Based Platform for River Investigation** *Jun. 2019 – Jun. 2021*
 - Developed a high-fidelity LiDAR simulation and analysis framework to derive optimal hardware specifications, validate dual-wavelength bathymetric sensing performance, and translate point-cloud-based error modeling into actionable embedded-system requirements for high-accuracy river and subsurface mapping.
- **3D Vehicle Shape Generation System for Automated Car Wash** *Jan. 2019 – Dec. 2020*
 - Developed a fast, sub-2 cm-accuracy 3D vehicle-profiling system for unmanned car-wash automation, integrating custom multi-LiDAR sensing, fused point-cloud reconstruction, and real-time TI-embedded processing to ensure reliable geometry extraction and collision-free operation.

PUBLICATIONS

- *Journal Articles*
- Younghun Byeon, Eunju Kim, **Hyeon Jun Lim**, Han Sol Kim, "Development of a Faster R-CNN-based Marine Debris Detection Model for an Embedded System," *Journal of Institute of Control, Robotics and Systems*, 27(12), 1038–1043 (KCI), 2021.
- Younghun Byeon, Minjae Kang, **Hyeon Jun Lim**, Han Sol Kim, "Development of a Three-Dimensional Indoor Position Estimation Algorithm for a Drone Using Stereo Vision and YOLOv3," *Journal of Institute of Control, Robotics and Systems*, 27(11), 845–850 (KCI), 2021.
- **Hyeon Jun Lim**, Young Hoon Joo, Jin Bae Park, "A Relaxed Stabilization Condition for Discrete T-S Fuzzy Model under Imperfect Premise Matching," *Journal of Korean Institute of Intelligent Systems*, 27(1), 59–64 (KCI, IF 0.47), 2017.
- *International Conferences*
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "Fuzzy Observer Design and Its Experimental Validation Using the Fuzzy Lyapunov Function," 17th International Conference on Control, Automation and Systems (ICCAS), Jeju, Korea, 2017.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "Fuzzy Lyapunov Function-based H_∞ Fuzzy Filter Design for Sampled-data Systems under Imperfect Premise Matching," SICE Annual Conference, Kanazawa, Japan, 2017.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "The Stabilization Condition of Continuous Affine Fuzzy Systems Under Imperfect Premise Matching," 16th International Conference on Control, Automation and Systems (ICCAS), Gyeongju, Korea, 2016.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "The Fuzzy Lyapunov Function-based Relaxed Stabilization of Output-feedback Fuzzy Control Systems Under Imperfect Premise Matching," International Conference on Fuzzy Theory and Its Applications (iFuzzy), Taichung, Taiwan, 2016.
- *Domestic Conferences*
- **Hyeon Jun Lim**, Seungho Baeg, "An Improvement Method for Global Mapping of Point Cloud Data Through the Statistical Outlier Removal," 16th Korea Robotics Society Annual Conference (KRoC), Pyeongchang, Korea, 2021.

- **Hyeon Jun Lim**, Seungho Baeg, “A Time Compensation Method for a Scanning LiDAR Sensor via Hermite Interpolation,” KIIS Autumn Conference, 30(2), 2020.
- **Hyeon Jun Lim**, Seungho Baeg, “A Calibration Method for a 2D LiDAR Sensor and a Linear Encoder System Using Dihedral Angle,” KIEE Conference on Information and Control Systems (CICS), Jeju, Korea, 2020.
- **Hyeon Jun Lim**, Seungho Baeg, Jae-Hak Kim, Jae-Sung Jeong, Changsung Kim, “Analysis of Design Parameters and Performance Simulation for a Bathymetric LiDAR Sensor,” KSCE Convention, Drone Platform-based River Survey Special Session, 2020.
- Seungho Baeg, Beomjin Hong, Suhyun Na, **Hyeon Jun Lim**, “Korea Dataset for Autonomous Solution and the Acquisition System,” KSAE Autumn Conference, 2018.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, “Fuzzy Output-feedback Controller Design for T-S Fuzzy Models based on Improved Fuzzy Lyapunov Functions under Imperfect Premise Matching,” ICROS Annual Conference, Sokcho, Korea, 2017.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, “Output-feedback Controller Design for T-S Fuzzy Models based on Fuzzy Lyapunov Functions: Guaranteed-cost Approach,” KIIS Spring Conference, 27(1), 2017.
- **Hyeon Jun Lim**, Jin Bae Park, Taek Gyu Song, Young Hoon Joo, “The Relaxed Stabilization Condition for the Affine T-S Fuzzy Systems via Improved Fuzzy Lyapunov Functions: Guaranteed Cost Approach,” KIEE Summer Conference, 48th, 2017.
- **Hyeon Jun Lim**, Chang-Gil Lee, Jin Bae Park, Young Hoon Joo, “Fuzzy Output-feedback Control Design for T-S Fuzzy Models under Imperfect Premise Matching,” KIEE Conference on Information and Control Systems (CICS), 2016.
- **Hyeon Jun Lim**, Han Sol Kim, Jin Bae Park, Myoung Hwan Tak, Young Hoon Joo, “Stabilization for the T-S Fuzzy Systems Using the Line-integral Fuzzy Lyapunov Function,” KIIS Spring Conference, 26(1), 2016.
- **Hyeon Jun Lim**, Young Hoon Joo, Jin Bae Park, “A Relaxed Stabilization Condition for Discrete T-S Fuzzy Model under Imperfect Premise Matching,” KIIS Autumn Conference (Best Paper Award; later extended into the 2017 JKIIIS article), 2016.

PATENTS

- “Apparatus and Method for Processing Trajectory Data,” KR 10-2026-0139322 (Application), Filed Jul 28, 2026.
- “System For Processing Data Required For Object Detection Based On Artificial Intelligence and Multiple Autonomous Driving Sensors and Method Implementing The Same,” KR 10-2025-0090242 (Application), Filed Jul 4, 2025. Also filed: US 19/415,494 · CN 202511840140.4.
- “Apparatus For Controlling Vehicle and Method Thereof,” KR 10-2025-0044855 (Application), Filed Apr 7, 2025. Also filed: US 19/391,095 · DE 10 2025 146 420.0 · CN 202511797806.2.
- “System and Method For Calibrating 3D Sensor Coordinate Based On Scene Flow Network,” KR 10-2025-0042966 (Application), Filed Apr 2, 2025. Also filed: US 19/406,040 · CN 202511849897.X.
- “System and Method For Adaptive Calibration For Autonomous Driving,” KR 10-2025-0042965 (Application), Filed Apr 2, 2025. Also filed: US 19/409,994 · CN 202511849891.2.
- “3D Vehicle Shape Data Acquisition Method and System for Automated Car Wash Equipment,” KR 10-2504548 (Registered), Filed Dec 7, 2020 · Registered Feb 23, 2023.
- “3D Vehicle Shape Data Acquisition Method and System for Automated Car Wash Equipment, Rotating LiDAR Configuration,” KR 10-2504547 (Registered), Filed Dec 7, 2020 · Registered Feb 23, 2023.

SOFTWARE COPYRIGHTS

- **Hyeon Jun Lim**, Seungho Baeg, “Laser Beam Characteristics Evaluation Program,” Korea Copyright Commission, C-2021-008734, Feb 2021.
- **Hyeon Jun Lim**, Seungho Baeg, “LiDAR Sensor Data Viewer Program,” Korea Copyright Commission, C-2020-052146, Dec 2020.
- **Hyeon Jun Lim**, Seungho Baeg, “LAS File Conversion Program,” Korea Copyright Commission, C-2020-052137, Dec 2020.

- Seungho Baeg, **Hyeon Jun Lim**, “NovAtel GNSS Data Reception Firmware for TI ARM Processor,” Korea Copyright Commission, C-2020-052144, Dec 2020.
- Seungho Baeg, **Hyeon Jun Lim**, “Range Data Acquisition Firmware for DSP Processor,” Korea Copyright Commission, C-2020-052143, Dec 2020.
- Seungho Baeg, **Hyeon Jun Lim**, “PRU Firmware for Motor Speed Control and Rotation Angle Detection on TI DSP,” Korea Copyright Commission, C-2020-052142, Dec 2020.
- Seungho Baeg, **Hyeon Jun Lim**, “CAN and Serial Communication API Driver for TI DSP,” Korea Copyright Commission, C-2020-052141, Dec 2020.
- Seungho Baeg, **Hyeon Jun Lim**, “Epson IMU Data Reception Firmware for PRU Processor,” Korea Copyright Commission, C-2020-052140, Dec 2020.